Supplementary Material: Identification Theory for Format-Confounded Behavioral Estimands

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This supplement contains the full statements and proofs of the identification results summarized in Appendix A of the main paper ("Discriminating Real Post-Training Displacement from Its Confounds in LLM Price Negotiation"). It is self-contained and compiles independently of the main document. As the main paper states in four places, *none of the empirical findings depend on these results*: they port classical misclassification and censoring identification to the format-confound setting and isolate the one genuinely new object (history-correlated differential failure). Section and equation numbers here are internal to this supplement; references to the main paper are given in prose.

1 Identification theory

This appendix states and proves the identification results the main text ports; *none of the empirical findings in the paper depend on them.* As a reader's guide: Appendices 1.1–1.3 port standard misclassification and censoring identification—the load-bearing payoff being the (E)/(X) characterization and the partial-identification diameter—and the one genuinely new object, history-correlated differential failure, lives in Appendices 1.4–1.6. The estimand is strategic competence C (the intended policy π_C); the nuisance is output-format fidelity F (per-action parse probability $q_F(a, h)$); the confound is that, when a harness defaults a parse-failed turn to an in-distribution action, a format failure is scored as a strategic failure. The static theory (Appendices 1.1–1.3) is a faithful port of the econometrics of misclassification and censoring [Bollinger, 1996, Mahajan, 2006, Hu, 2008, Manski, 2003]; the horizon analysis (Appendices 1.4–1.6) is where the one genuinely new object lives—history-correlated differential failure—together with a rigorous over-determination result whose degenerate locus is named

exactly. We mark each result for whether it is load-bearing and whether it is proved or sketched.

1.1 Model

Game, competence, channel. At turn t the agent observes a history h_t and selects an action a_t from an action space A; play produces a trajectory τ. Competence C parameterizes the intended policy $\pi_C(a \mid h)$, and the estimand is a scalar signature $\theta = \mathbb{E}_{\pi_C}[\phi]$ for a feature $\phi : A \to \mathbb{R}$. We assume $C \mapsto \pi_C$ is injective, which defines what "recovering C" means. The format channel F governs emission: an intended action a at h parses successfully with probability $q_F(a, h)$, and we adopt the baseline *faithful-when-parseable* (when the string parses, the parser returns exactly a; the confound runs through failures, not misreads—relaxing this never helps).

The garbling chain. Intended actions are garbled into effective actions that drive the dynamics and the counterpart (Figure 1):

$$\tilde{a}_t = \begin{cases} a_t & \text{w.p. } q_F(a_t, h_t), \\ \delta(h_t) & \text{w.p. } 1 - q_F(a_t, h_t), \end{cases} \qquad C \longrightarrow a_{1:T} \xrightarrow{F} \tilde{a}_{1:T} \longrightarrow \tau \xrightarrow{\sigma} S, \tag{1}$$

where $\delta(h)$ is the harness default on a parse failure. F is a Blackwell garbling of the action sequence; by the data-processing inequality the protocol can only destroy information about C relative to the channel. A protocol is $P = (\rho, \delta, \sigma)$: a parser $\rho : \text{string} \to A \cup \{\bot\}$, a default rule δ, and a scorer σ. The *format-oracle* protocol P^{or} has the same scorer but a parser that never fails (it reads the intended a_t directly); it isolates the incremental damage caused by format.

The two structural properties.

Definition 1 ((E): failures exposed)**.** S determines, per turn, the parse-failure indicator $\mathbf{1}[\rho(\cdot) = \bot]$; equivalently δ routes failures to a symbol $\bot^\star \notin A$ that σ treats separably from every intended action. A fixed additive penalty folded into a scalar score does *not* qualify. The negation of (E) is *Defect A*.

Definition 2 ((X): action-exogeneity)**.** For all F and h, the failure probability $q_F(a, h)$ is constant in a over $\operatorname{supp} \pi_C(\cdot \mid h)$; write it $q_F(h)$. The negation of (X) is *Defect B*.

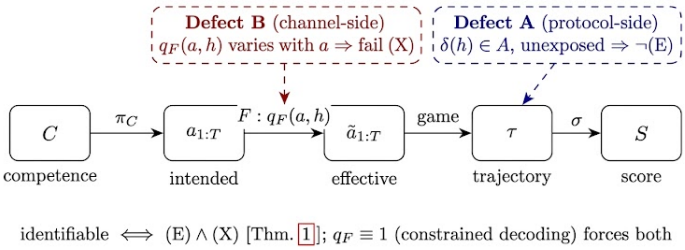


Figure 1: The garbling chain of (1). Competence C induces intended actions $a_{1:T}$; the format channel F garbles them into effective actions $\tilde{a}_{1:T}$ (an intended action that fails to parse is replaced by the harness default $\delta(h)$); the game produces a trajectory τ, scored as S. The two confounds attach to different links: *Defect B* is action-dependent failure on the F link (negating exogeneity (X)); *Defect A* is the unexposed default collapsing failures onto a real action before scoring (negating exposure (E)). Identifiability requires both (E) and (X); forcing $q_F \equiv 1$ (constrained decoding) is the one intervention that guarantees them, at the cost of the grammar projection quantified in the main paper (format wedge, Discriminator 1).

Definition 3 (Format-identifiability)**.** C is format-identifiable under P over $C \times F$ iff for all C_1, C_2 distinguishable under P^{or} and all F_1, F_2, $\text{Law}_{C_1,F_1}(S) = \text{Law}_{C_2,F_2}(S) \Rightarrow C_1 = C_2$.

1.2 Static non-identifiability and its characterization

Lemma 1 (Non-identifiability under a colliding default)**.** *Let $h^\star$ be reached with positive probability, let $a^\circ := \delta(h^\star) \in A$, and suppose (i) σ distinguishes a° at $h^\star$ from some alternative a', and (ii) $a^\circ \in \operatorname{supp} \pi_C(\cdot \mid h^\star)$ for some admissible C. Then C is not format-identifiable.*

Proof. Fix $\beta \in (0,1)$ and let the two agents differ only at $h^\star$. Agent (C_1, F_1) intends a' with probability 1 at $h^\star$ but with format failure $q_{F_1}(a', h^\star) = 1-\beta$, so its effective action at $h^\star$ is a° w.p. β and a' w.p. $1-\beta$. Agent (C_2, F_2) intends a° w.p. β and a' w.p. $1-\beta$, with $q_{F_2} \equiv 1$; its effective action is the same mixture. Outside $h^\star$ set $C_1 = C_2$ and $q_{F_1} = q_{F_2} = 1$. The two induce identical effective-action laws at every history, hence identical τ-laws (the counterpart sees only effective actions), hence identical $\text{Law}(S)$. But under P^{or} the intended policies differ, $\pi_{C_1}(\cdot \mid h^\star) = \delta_{a'} \neq \beta\delta_{a^\circ} + (1-\beta)\delta_{a'} = \pi_{C_2}(\cdot \mid$

$h^\star$), and by (i) the oracle distinguishes them. The construction holds on the whole interval $\beta \in (0,1)$, so the confounded region has positive measure. □

Theorem 1 (Characterization)**.** *C is format-identifiable under P over $C \times F$ iff (E) and (X) both hold (with $q_F \equiv 1$ the degenerate case satisfying both).*

Proof. Sufficiency. Under (E), the per-turn observation at h is the full distribution over $A \cup \{\bot^\star\}$: $\nu_h(a) = \pi_C(a \mid h)\, q_F(a,h)$ for $a \in A$, and $\nu_h(\bot^\star) = \sum_a \pi_C(a \mid h)(1 - q_F(a,h))$. Under (X), $q_F(a,h) = q_F(h)$, so $\nu_h(a) = q_F(h)\, \pi_C(a \mid h)$ and

$$\pi_C(a \mid h) = \frac{\nu_h(a)}{\sum_{a' \in A} \nu_h(a')} \qquad \text{for every } a, h. \tag{2}$$

By injectivity of $C \mapsto \pi_C$, C is recovered. (Faithful-when-parseable is used here; a misread channel replaces $\nu_h(a)$ by a mixture through a confusion matrix, identifiable only if that matrix is known or invertible—it never *adds* identifiability.)

Necessity of (E). Suppose (X) holds but (E) fails, so failures route to $a^\circ = \delta(h^\star) \in A$. Lemma 1 applies, and its construction respects (X) because $q_{F_1}(a', h^\star) = 1 - \beta$ is action-constant on the singleton support $\{a'\}$.

Necessity of (X). Suppose (E) holds but (X) fails: a reachable $h^\star$ has two actions a_0, a_1 with unequal failure probabilities. With $\bot$ exposed, the observable at $h^\star$ is $(\pi_C(a) q_F(a))_a$ together with the failure mass. The two configurations

	$\pi(a_0)$	$q(a_0)$	$q(a_1)$	$\nu(a_0)$	$\nu(a_1)$	$\nu(\bot^\star)$
(C_1, F_1)	0.8	0.5	1.0	0.4	0.2	0.4
(C_2, F_2)	2/3	0.6	0.6	0.4	0.2	0.4

produce the identical observable $(0.4, 0.2, 0.4)$—the failure indicator is fully exposed—yet $\pi_{C_1}(a_0) = 0.8 \neq 2/3 = \pi_{C_2}(a_0)$. The example is an open set, hence non-pathological. □

Corollary 1 (The disjunction is false)**.** *The natural conjecture—"C identifiable iff $\bot$ is exposed or the channel is equalized"—is incorrect. The necessity-of-(X) construction satisfies "$\bot$ exposed" yet is non-identifiable. The correct criterion is the conjunction (E)∧(X): exposing $\bot$ removes Defect A but is blind to Defect B; one additionally needs the parseable subset to be unbiased for C, i.e. non-differential failure.*

Corollary 2 (No protocol fix in the differential regime)**.** *Let F_{diff} contain any channel with $q_F(a_0, h^\star) \neq q_F(a_1, h^\star)$ for some reachable $h^\star$. Then no protocol P that admits parse failures is format-identifiable over $C \times F_{\text{diff}}$; the only identifying protocols force $q_F \equiv 1$ (constrained decoding).*

Proof. Any P observes a function of the effective-action law plus, at best, the failure indicator—i.e. a function of $(\pi_C(a)q_F(a))_a$ and $\nu(\perp^\star)$. The two-row table of Theorem 1 matches that entire object across distinct C for an open set of parameters, for every such P. The only way to observe more is to read the intended action directly, which requires the parser never to fail. □

1.3 Quantitative partial identification

When non-identifiable, the damage is an interval. Work at a single history h in the Defect-A regime with an exogenous channel: $\mu(a) = q\,\pi_C(a) + (1-q)\mathbf{1}[a = a^\circ]$, with q and π_C both unknown, and the signature $\theta = \sum_a \pi_C(a)\phi(a)$.

Theorem 2 (Identified-set diameter, single history)**.** *The identified set for θ is an interval with*

$$\operatorname{diam}\Theta_{\text{id}} = \underbrace{\frac{\mu(a^\circ)}{1-\mu(a^\circ)}}_{\textit{collision mass}} \cdot \underbrace{\big|\,\mathbb{E}_\mu[\phi] - \phi(a^\circ)\,\big|}_{\textit{collision geometry}}, \tag{3}$$

and both endpoints are attained.

Proof. Given μ, the consistent (π_C, q) are exactly those with $\pi_C(a) = \mu(a)/q$ for $a \neq a^\circ$ and $\pi_C(a^\circ) = (\mu(a^\circ) - (1-q))/q$. Non-negativity of $\pi_C(a^\circ)$ forces $q \in [1-\mu(a^\circ), 1]$, and $\sum_a \pi_C(a) = 1$ holds automatically. Writing $m := \mathbb{E}_\mu[\phi]$,

$$\theta(q) = \sum_a \frac{\mu(a)}{q}\phi(a) - \frac{1-q}{q}\phi(a^\circ) = \phi(a^\circ) + \frac{m - \phi(a^\circ)}{q},$$

continuous and monotone in q. Hence the identified set is the interval between $\theta(1) = m$ and $\theta(1-\mu(a^\circ))$, and $\operatorname{diam} = |m-\phi(a^\circ)|\big(\frac{1}{1-\mu(a^\circ)} - 1\big) = \frac{\mu(a^\circ)}{1-\mu(a^\circ)}|m-\phi(a^\circ)|$. The extremes $q \in \{1-\mu(a^\circ), 1\}$ are the Lemma 1 construction. □

Instantiation (auto-reveal-on-missing-tag). An agent emits its action inside required tags; on a missing tag the harness substitutes a default that capitulates, scored as a bad outcome. Model $\delta(h) = a^\circ$ = "concede full surplus", ϕ = captured surplus, so $\phi(a^\circ) = 0$ and (3) gives diam $= \frac{\mu(a^\circ)}{1-\mu(a^\circ)}\,|\bar{S}|$. If 30% of trajectories hit the worst outcome ($\mu(a^\circ) = 0.3$) and mean captured surplus is $\bar{S} = 0.40$, the competent surplus is identified only to a band of width $\frac{0.3}{0.7} \cdot 0.40 \approx 0.17$, i.e. $[0.40, 0.57]$. Seventeen surplus-points of the apparent strategic spread are unidentified—attributable to the format default, not to strategy.

The binary impossibility (Corollary 2) is too coarse for practice; we need the degradation curve under a *bounded* violation of (X). Suppose $\perp$ is exposed but the channel is still differential, with budget $\eta := \sup_{a,a',h} |q_F(a,h) - q_F(a',h)|$. (From bargaining logs we never observe $q_F(a,h)$ alone; η is recovered by a *format probe*—teacher-force each action and measure per-action parse success.) Write $\bar{\nu} := \nu/Q$ with $Q := \sum_a \nu(a)$ and $\theta_0 := \mathbb{E}_{\bar{\nu}}[\phi]$.

Theorem 3 (η-robust diameter)**.** *Fix a history h with $\perp$ exposed and channel spread $\leq \eta$. The identified set for θ is an interval, and*

$$\operatorname{diam}(\Theta_\eta) = \frac{\eta}{Q}\,\mathbb{E}_{\bar{\nu}}\big|\phi - \theta_0\big| + O(\eta^2) = \frac{\eta}{Q}\,\mathrm{MAD}_{\bar{\nu}}(\phi) + O(\eta^2). \tag{4}$$

The leading term is exact. For all $\eta < Q$ there is the two-sided bound $\frac{\eta}{Q}\mathrm{MAD}_{\bar{\nu}}(\phi) \leq \operatorname{diam}(\Theta_\eta) \leq \frac{2\eta}{Q-\eta}\mathrm{MAD}_{\bar{\nu}}(\phi)$.

Proof. Under (E) the observable is $\nu(a) = \pi_C(a)q(a)$ plus $\nu(\perp^\star) = 1 - Q$. Any consistent (π, q) has $\pi(a) = \nu(a)/q(a)$, so the nuisance is the success vector $q(\cdot)$ subject to $\sum_a \nu(a)/q(a) = 1$, $\max_a q(a) - \min_a q(a) \leq \eta$, $q(a) \in [\nu(a), 1]$, with $\theta(q) = \sum_a \nu(a)\phi(a)/q(a)$. All $q(a)$ lie in a window $[\underline{q}, \underline{q} + \eta]$; normalization forces $\underline{q} \leq Q \leq \underline{q} + \eta$, so $|Q - q(a)| \leq \eta$ for every a. The normalization also gives the exact recentering

$$\theta - \theta_0 = \sum_a \nu(a)\big(\phi(a) - \theta_0\big)\Big(\tfrac{1}{q(a)} - \tfrac{1}{Q}\Big). \tag{5}$$

For the global bound, $|\frac{1}{q(a)} - \frac{1}{Q}| = \frac{|Q-q(a)|}{q(a)Q} \leq \frac{\eta}{(Q-\eta)Q}$, whence $|\theta - \theta_0| \leq \frac{\eta}{Q-\eta}\mathrm{MAD}_{\bar{\nu}}(\phi)$ and diam $\leq 2\sup|\theta - \theta_0|$. For the leading order write $q(a) = Q + \varepsilon(a)$, $|\varepsilon(a)| \leq \eta$, with $\sum_a \nu(a)\varepsilon(a) = 0$ to first order; then $\frac{1}{q(a)} - \frac{1}{Q} = -\varepsilon(a)/Q^2 + O(\eta^2)$ and (5) is linear in ε over the spread-η box. The extreme vertices set $\varepsilon(a) = \eta\,\mathbf{1}[\pm(\phi(a) - \theta_0) > 0]$; since the positive and negative parts of $\phi - \theta_0$ carry equal ν-mass, summing the two extremes gives diam $= \frac{\eta}{Q^2}\sum_a \nu(a)|\phi(a) - \theta_0| + O(\eta^2) = \frac{\eta}{Q}\mathrm{MAD}_{\bar{\nu}}(\phi) + O(\eta^2)$. □

Binary closed form (tightness certificate). For $A = \{0, 1\}$ with $p = \nu_0/Q$, $\theta = \phi_1 + \pi_0(\phi_0 - \phi_1)$ is affine in $\pi_0 = \nu_0/q_0$, and $g(\pi_0) := q_0 - q_1 = \nu_0/\pi_0 - \nu_1/(1-\pi_0)$ is strictly decreasing. The constraint $|g| \le \eta$ gives roots $\pi_0^{\pm} = p \pm \frac{\nu_0\nu_1}{Q^3}\eta + O(\eta^2)$, so diam $= |\phi_0 - \phi_1|(\pi_0^+ - \pi_0^-) = \frac{2\nu_0\nu_1}{Q^3}\eta|\phi_0 - \phi_1| + O(\eta^2) = \frac{\eta}{Q}\mathrm{MAD}_{\bar\nu}(\phi)$, matching (4) since $\mathrm{MAD}_{\bar\nu}(\phi) = 2p(1-p)|\phi_0 - \phi_1|$.

1.4 The horizon: phantom contamination and recovery

In multi-turn play a parse failure at turn t injects $\delta(h_t)$ into the history both agents condition on thereafter, raising the worry that information degradation compounds over the horizon. Under the fine-observation model that worry assumed, it is a phantom.

Lemma 2 (Contamination is a phantom under (E)$\wedge$(X))**.** *Suppose (E), (X) with $q := \inf_h q_F(h) > 0$. Then π_C is identified on its entire intended-reachable support, and the intended-path signature is point-identified:* $\mathrm{diam}(\Theta) = 0$ *for every horizon T.*

Proof. Let h be intended-reachable. The fully-clean event reaching h (no failure on the way) is observable by (E) and has probability $\ge q^{|h|} > 0$; on it the effective history equals h. Condition further on "no failure at the next turn" (observable). By (X), failure is independent of the intended action given h, so this conditioning does not tilt the intended action distribution; and faithful-when-parseable makes the observed next action equal the intended one. Hence the law of the observed next action given (clean-h, no-failure-next) is exactly $\pi_C(\cdot \mid h)$, at every intended-reachable h. □

So under fine observation the estimand is recovered by *discarding* failed trajectories, at a sample-size cost q^{-T}: compounding is statistical, not identificational. The diameter is genuine only under coarse outcome scoring, which we now make exact.

Hold-or-Fold. Horizon T, pot R. Each round the focal agent emits $a_t \in \{H, L\}$ (H = hold/demand, L = fold), with $\phi(H) - \phi(L) = R$. A parse failure substitutes $\delta = L$. The channel is non-differential with per-round failure $r = 1 - q$ (intended $H \to$ effective L w.p. r; intended $L \to L$ always; a defaulted L is indistinguishable from an intended L). The counterpart *Tit-for-the-pot* yields the whole pot iff the focal agent effectively holds every round; on the first effective L it seizes the pot. The aggregate score is $S = \mathbf{1}[\tilde a_{1:T} = H^T]$. The estimand is the intended hold-through rate $\Pi := \mathbb{P}(\sigma > T)$ (σ = intended first-fold time), with surplus signature $\theta = R\,\Pi$.

Proposition 1 (Coarse-scoring diameter)**.** *The only observable is the win rate $p = \Pi\, q^T$, the identified set is $\theta \in [Rp,\, R]$ with* $\operatorname{diam}(\Theta) = R(1-p)$*, and both endpoints are attained by genuinely distinct agents.*

Proof. Only an intend-hold-all and never-corrupted trajectory wins, so $p = \Pi q^T$. Given p, the class admits $\Pi = p/q^T$ for any $q \in [p^{1/T}, 1]$ (to keep $\Pi \le 1$), so $\theta = R\Pi$ sweeps $[Rp, R]$ continuously. The endpoints are realized by Hi (mediocre hawk, clean format: $q = 1$, $\Pi = p$, $\theta = Rp$) and Lo (perfect hawk, broken format: $q = p^{1/T}$, $\Pi = 1$, $\theta = R$); both emit $S \sim \text{Bernoulli}(p)$, so $\text{Law}(S)$ is identical while $\theta_{\text{Lo}} - \theta_{\text{Hi}} = R(1-p)$. □

For the full-hawk signature ($\Pi = 1$, $p = q^T$), $\operatorname{diam}(\Theta) = R(1-q^T) \to R$ as $T \to \infty$: outcome-style scoring of long bargains is asymptotically fully non-identifying for intended competence. (The conjunctive surplus signature gives diameter $R(1-q^T)$, not the additive time-average $(1-\bar{c}_T)R$; since $q^T \le \bar{c}_T := \frac{1}{T}\sum_t q^t$, the conjunctive quantity is the larger, and correct, one.)

1.5 The differential channel and the reversal

Run a differential channel (q_H, q_L), $|q_H - q_L| \le \eta$, through Hold-or-Fold; σ is the intended fold time and $G(k) := \mathbb{P}(\sigma > k)$ the intended survival ($G(0) = 1$), estimand $\theta = R\,G(T)$. The realistic case is $q_H < q_L$: a hold carries more structure (justification, conditions) than a fold, so holds fail to parse more often and the default silently converts them to folds. The counterpart seizes the pot on the first effective L, giving outcome categories

$$p_{\text{win}} = G(T)\, q_H^T, \tag{6}$$

$$\ell_k = \big(G(k-1) - G(k)\big) q_H^{k-1} q_L \qquad \text{(terminal parsed-}L\text{ at } k\text{)}, \tag{7}$$

$$b_k = q_H^{k-1}\big[G(k)(1-q_H) + (G(k-1) - G(k))(1-q_L)\big] \qquad \text{(terminal } \perp \text{ at } k\text{)}. \tag{8}$$

Proposition 2 (Hidden $\perp$ erases q_L and the horizon)**.** *Under Defect A (a defaulted L indistinguishable from an intended L), the observable law depends only on (G, q_H)—q_L drops out—and the identified set is $\theta \in [Rp, R]$,* $\text{diam} = R(1-p)$*, for any channel and any horizon.*

Proof. With $\perp$ hidden, ℓ_k and b_k merge into "observed L at k,"

$$m_k = \ell_k + b_k = q_H^{k-1}\big[(G(k-1) - G(k))\underbrace{(q_L + 1 - q_L)}_{=1} + G(k)(1-q_H)\big] = q_H^{k-1}\big(G(k-1) - q_H G(k)\big),$$

in which q_L has cancelled. The observables are $p_{\text{win}} = G(T)q_H^T$ and $\{m_k\}$, functions of (G, q_H) only; fixing q_H determines every $G(k)$ with $G(T) = p_{\text{win}}/q_H^T$, and $q_H \in [p_{\text{win}}^{1/T}, 1]$ makes $\theta = Rp_{\text{win}}/q_H^T$ sweep $[Rp, R]$. □

This is why *exposing* $\perp$ is the relevant lever: with $\perp$ hidden the richer trajectory buys nothing. With $\perp$ exposed at $T = 1$ (emissions read as $\{H, L, \perp\}$, $a = \pi q_H$, $b = (1-\pi)q_L$, $Q = a + b$), the identified set for π admitting $q_H = a/\pi \le 1$, $q_L = b/(1-\pi) \le 1$, $|q_H - q_L| \le \eta$ is a middle interval; at $\eta = 0$ it collapses to $\pi^\star = a/(a+b)$ (Lemma 2 recovered), and to leading order $\operatorname{diam}(\Theta_\eta) = R \cdot \frac{2ab}{(a+b)^3}\eta + O(\eta^2) = \frac{\eta}{Q}\text{MAD}_{\tilde{\nu}}(\phi)$—Theorem 3 re-derived bottom-up from a concrete channel.

1.6 Hardening Proposition 2: global identification and the degenerate locus

Whether the η-band of the previous subsection averages away over T splits on channel homogeneity. We give the rigorous over-determination result and characterize its failure set exactly.

Setup. Work inside the homogeneous-channel Hold-or-Fold model with $\perp$ exposed. The parameter is $\vartheta = (q, r, G(1), \dots, G(T))$ with $q := q_H \in (0, 1]$, $r := q_L \in [0, 1]$, $1 = G(0) \ge G(1) \ge \cdots \ge G(T) \ge 0$, and $\dim \Theta = T + 2$; the estimand is $\theta = R\,G(T)$. With $\perp$ exposed every game ends in one of $2T + 1$ outcomes, with probabilities (6)–(8) (now $q_H = q$, $q_L = r$ constant). Define the failure sets

$$\Theta_{\text{geo}} := \{\vartheta : G(k) = G(1)^k \ \forall k\} \text{ (geometric / constant-hazard)}, \qquad \Theta_{\text{deg}} := \Theta_{\text{geo}} \cup \{r = 0\}.$$

Θ_{geo} is cut out by the $T - 1$ equations $G(k) = G(1)^k$ $(k \ge 2)$, so it has codimension $T - 1$; $\{r = 0\}$ has codimension 1. Hence Θ_{deg} is closed with empty interior, and $\Theta \setminus \Theta_{\text{deg}}$ is open and dense.

Theorem 4 (Rigorous Proposition 2)**.** *Fix $T \ge 2$. (i) On $\Theta \setminus \Theta_{\text{deg}}$ the observation map Φ is injective: $q, r, G(1), \dots, G(T)$—hence θ—are point-identified by the closed form* (11) *below, for every admissible channel including a fully differential one. (ii) At every $\vartheta \in \Theta_{\text{geo}}$ with $0 < G(T) < G(1) < 1$, if the channel may be differential then θ is not point-identified; its identified set is the interval* (12)*. The same holds on $\{r = 0\}$. (iii)* $\operatorname{rank} D\Phi(\vartheta) = T + 2$ *iff $\vartheta \notin \Theta_{\text{deg}}$; on Θ_{deg},* $\operatorname{rank} D\Phi \le T + 1$ *with the explicit kernel* (13)*. The count $2T \ge T + 2$ ($T \ge 2$) is thus necessary but not sufficient: Θ_{deg} is exactly where the count holds but the rank collapses.*

Lemma 3 (Reduction)**.** *Define the decay products* $u_k := q^k G(k)$ *(*$u_0 = 1$*). With* $m_k := \ell_k + b_k$*,*

$$m_k = u_{k-1} - u_k, \qquad p_{\text{win}} = u_T, \qquad \ell_k = r\,u_{k-1} - \tfrac{r}{q}\,u_k. \tag{9}$$

Consequently the products are identified from data, $u_k = 1 - \sum_{j\le k} m_j$*; and writing* $\alpha := r$*,* $\beta := r/q$*, the residual content of* $\{\ell_k\}$ *is the linear system* $\ell_k = \alpha\,u_{k-1} - \beta\,u_k$*,* $k = 1, \dots, T$*.*

Proof. From (6)–(8), $m_k = q^{k-1}[d_k + G(k)(1-q)] = q^{k-1}[G(k-1) - qG(k)]$ with $d_k := G(k-1) - G(k)$; since $q^{k-1}G(k-1) = u_{k-1}$ and $q^kG(k) = u_k$ this is $u_{k-1} - u_k$. Summing telescopes to $u_k = 1 - \sum_{j\le k} m_j$, and $p_{\text{win}} = G(T)q^T = u_T$. For ℓ_k, $d_k q^{k-1} = u_{k-1} - u_k/q$, so $\ell_k = r(u_{k-1} - u_k/q) = \alpha u_{k-1} - \beta u_k$. The observables $(p_{\text{win}}, \ell, b)$ re-encode bijectively as $(\{u_k\}, \{\ell_k\})$ via $b_k = m_k - \ell_k$. □

The product relations are the Defect-A content: hidden-$\perp$ observables fix only the products $u_k = q^kG(k)$, never q and $G(k)$ separately. All identifying power beyond Defect A is in the split of m_k into ℓ_k and b_k.

Lemma 4 (Rank criterion)**.** *Let* $M \in \mathbb{R}^{T\times 2}$ *have rows* $(u_{k-1}, -u_k)$*. Then* $\operatorname{rank} M = 2$ *iff* $\{u_k\}$ *is non-geometric, iff* $\vartheta \notin \Theta_{\text{geo}}$*. The* 2×2 *minor on rows* $i < j$ *is*

$$u_i u_{j-1} - u_{i-1}u_j = q^{\,i+j-1}\big[G(i)G(j-1) - G(i-1)G(j)\big], \tag{10}$$

vanishing for all $i < j$ *iff* $\log G$ *is affine, i.e.* $G(k) = G(1)^k$*.*

Proof. The minor on rows i, j is $u_iu_{j-1} - u_{i-1}u_j$; substituting $u_k = q^kG(k)$ gives (10). The bracket is the discrete Wronskian of G; it vanishes for all $i < j$ iff $\log G$ is affine in k. M has rank < 2 iff all its 2×2 minors vanish. □

Proof of Theorem 4(i). Given the data we know $\{u_k\}$ and $\{\ell_k\}$ exactly (Lemma 3). Take $\vartheta \notin \Theta_{\text{deg}}$, so $r > 0$ and $\operatorname{rank} M = 2$ (Lemma 4). Then $M(\alpha, \beta)^\top = \ell$ has the unique solution $(\alpha, \beta) = (M^\top M)^{-1}M^\top \ell$, which equals $(r, r/q)$ since the data were generated by ϑ. Because $r > 0$, $\alpha > 0$ and $\beta > 0$, and

$$r = \alpha, \qquad q = \frac{\alpha}{\beta}, \qquad G(k) = \frac{u_k}{q^k}, \qquad \theta = R\,G(T) = \frac{R\,p_{\text{win}}}{q^T}. \tag{11}$$

Every coordinate of ϑ is a function of the observables, so Φ is injective on $\Theta \setminus \Theta_{\text{deg}}$. The argument uses no bound on $|q - r|$, so it holds for an arbitrarily differential channel. This is global and constructive—(11) is an algorithm—hence strictly stronger than local (Jacobian) identification. □

Proof of Theorem 4(ii). Take $\vartheta \in \Theta_{\mathrm{geo}}$, so $G(k) = c^k$ with $c := G(1) \in (0,1)$; then $u_k = (qc)^k$ is geometric and (6)–(8) collapse to functions of $w := qc$ and $s := r(1-c)$: $p_{\mathrm{win}} = w^T$, $\ell_k = s\, w^{k-1}$, $b_k = w^{k-1}(1-w-s)$, with $s = \ell_1$ and $w = p_{\mathrm{win}}^{1/T}$ observable. A vector $(q', r', G' = c'^{\bullet}) \in \Theta_{\mathrm{geo}}$ reproduces these iff $q'c' = w$ and $r'(1-c') = s$; a vector off Θ_{geo} cannot (geometric u' forces $G' \in \Theta_{\mathrm{geo}}$). Solving, $q' = w/c'$, $r' = s/(1-c')$; the constraints $q', r' \le 1$ read $c' \in [w, 1-s]$, a non-degenerate interval whenever $(q,r) \neq (1,1)$. Along it $\theta' = R\, c'^T$ is strictly increasing, so

$$\Theta_{\mathrm{id}}(\theta) = \big[\, R\, p_{\mathrm{win}},\ R\,(1-\ell_1)^T \,\big], \qquad \mathrm{diam} = R\big[(1-\ell_1)^T - p_{\mathrm{win}}\big]. \tag{12}$$

Exposing $\perp$ *does* help even here—it pins the top at $R(1-\ell_1)^T$ rather than the Defect-A value R—but does not close the band. The non-identification is driven by differential freedom: along the fiber $q' = w/c'$ and $r' = s/(1-c')$ move oppositely. If additionally $\eta = 0$ (so $q' = r'$), then $q'c' = w$ and $q'(1-c') = s$ sum to $q' = w+s$, $c' = w/(w+s)$: a single point. On $\{r = 0\}$, $\ell_k \equiv 0$ carries nothing, the data are only $\{u_k\}$, and $\theta = R\, u_T / q'^T$ varies over the Defect-A interval. □

Proof of Theorem 4(iii). Off Θ_{deg} the inverse (11) is a composition of rational maps, smooth there, so by the inverse function theorem $D\Phi$ has full column rank $T+2$. On Θ_{geo} (interior, $r > 0$), $\Phi = \Psi \circ (w, s)$ for a fixed smooth Ψ; the curve $\gamma(c') = (w/c',\ s/(1-c'),\ (c'^k)_{k=1}^T) \in \Theta_{\mathrm{geo}}$ holds (w,s) fixed, so $\Phi(\gamma)$ is constant and $D\Phi\, \dot\gamma = 0$, with tangent

$$v = \Big(\underbrace{-q/c}_{\partial q},\ \underbrace{r/(1-c)}_{\partial r},\ \underbrace{1, 2c, 3c^2, \ldots, Tc^{T-1}}_{\partial G(k) = kc^{k-1}}\Big) \in \ker D\Phi(\vartheta). \tag{13}$$

So $\operatorname{rank} D\Phi \le T+1$. On $\{r = 0\}$ (with G non-geometric) the curve $q' \mapsto (q', 0, (u_k/q'^k)_k)$ gives kernel $(1, 0, (-kG(k)/q)_k)$. □

The signal is the fold-hazard variation; the $T = 2$ certificate. Identification runs entirely on the minors (10), i.e. on the log-concavity defects $G(i)G(j-1) - G(i-1)G(j)$, which vanish exactly when the hazard is constant. At $T = 2$, $M = \left(\begin{smallmatrix} 1 & -u_1 \\ u_1 & -u_2 \end{smallmatrix}\right)$ and

$$\det M = u_1^2 - u_2 = q^2\big(G(1)^2 - G(2)\big) = q^2\, G(1)\big[\, G(1) - \mathbb{P}(\sigma > 2 \mid \sigma > 1) \,\big],$$

the change in conditional survival from round 1 to round 2. The recovery $q = \frac{u_1\ell_2 - u_2\ell_1}{\ell_2 - u_1\ell_1}$, $r = \frac{u_1\ell_2 - u_2\ell_1}{u_1^2 - u_2}$, $\theta = R\, u_2/q^2$ has denominators $\propto \det M$, so the inversion is ill-conditioned as $\vartheta \to \Theta_{\mathrm{geo}}$ (condition number $\sim 1/|G(1)^2 -$

$G(2)|$): the measure-zero failure set is the centre of a *thick* region of weak identification. A full-hold relation $b_2/b_1 = q$ does not certify identification in general: full-hold ($G \equiv 1$) is geometric, hence *on* Θ_{geo}, so that relation is a degenerate witness. The honest general certificate is $\det M = q^2(G(1)^2 - G(2)) \neq 0$ (non-constant fold-hazard), which is what (11) uses.

Placement and the one new object. On Θ_{geo}, $u_k = (qc)^k$ is a single geometric decay and the parsed-fold signal $\ell_k = s\, w^{k-1}$ shares the same rate $w = qc$, so it adds one number, not per-round information—exactly the obstruction in repeated-measurement deconvolution [Kotlarski, 1967, Hu and Schennach, 2008], with Θ_{geo} its exponential-family kernel. The horizon "reversal" is that machinery in discrete dress; our contribution is naming the locus. The genuinely new object remains the *inhomogeneous* breakdown—history-correlated $q_{H,t}\!\downarrow$ aliased to $G(k)\!\downarrow$—which this homogeneous analysis does not touch, which no marginal format probe can detect (the probe returns $\bar{q}_H$ averaged over histories), and which is precisely the kernel the firing test (Discriminator 2 of the main paper) probes empirically: absent across families on the synthetic probe and at every SFT/DPO checkpoint, and present only at the Tülu-RLVR checkpoint on realized decisive histories.

References

C. R. Bollinger. Bounding mean regressions when a binary regressor is mismeasured. *Journal of Econometrics*, 73(2):387–399, 1996.

Y. Hu. Identification and estimation of nonlinear models with misclassification error using instrumental variables: A general solution. *Journal of Econometrics*, 144(1):27–61, 2008.

Y. Hu and S. M. Schennach. Instrumental variable treatment of nonclassical measurement error models. *Econometrica*, 76(1):195–216, 2008.

I. Kotlarski. On characterizing the gamma and the normal distribution. *Pacific Journal of Mathematics*, 20(1):69–76, 1967.

A. Mahajan. Identification and estimation of regression models with misclassification. *Econometrica*, 74(3):631–665, 2006.

C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.